

Amber Li

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY **Expected May 2026**
Bachelor of Science, Electrical and Computer and Engineering, Computer Science Minor
GPA: 3.79; Dean's List Fall 2022 - Spring 2025
Cornell ECE Early Meng Program (+1 Semester) **Expected December 2026**

Relevant Courses: Intro to Microelectronics, Nanofabrication and Characterization of Electronics, Machine Learning, Foundations of Robotics, Signals and Systems, VLSI, Embedded Systems, Computer Architecture, Analog IC Design, Hardware Acceleration via FPGA, Complex Digital ASIC Design.

PROFESSIONAL EXPERIENCE

GE Aerospace Internship – Engine Engineering **May 2026-Aug 2026**

Plug Power Internship, Slingerlands, NY, *Electrical Engineer* **May 2025-Aug 2025**

- Designed, assembled, and tested a custom PCB using Altium to test a specific functionality of the fuel stack.
- Implemented a microcontroller to receive and transmit CAN, LIN, and SENT messages for facilitating the cross communication of different sensors.
- Assisted engineers in electrical debugging of motor systems.
- Gained experience using electrical measuring equipment, embedded software, circuit debugging, harness design, and manual PCB assembly.

Tokyo Electron Internship, Albany, NY, *Process Engineer (R&D)* **May 2023-Aug 2023**

- Fab work: Ran wafers through measurement tools and etch tools.
- Performed SEM sample cleaving, preparation, and imaging.
- Presented weekly project updates at meetings utilizing Microsoft applications, OriginLab, and other data visualization tools.
- Enabled engineers to achieve daily turnaround time for data collection and analysis on customer projects.

Cornell Nexus Project Team, Cornell University, *Electrical Subteam Lead* **Feb 2023-Present**

- Building an autonomous robot to traverse beaches and filter out microplastics from the sand.
- Implemented break beam sensors for robot's microplastic collection bucket, wrote Python code to detect bucket fill levels with ability to incorporate multiple sensors for expansion.
- Set up independent base-station and rover GPS-RTK modules to generate our own correction data for robot's positioning adjustments, researching wireless transmission of GPS data between modules.
- Collaborate with members from other subteams on interdisciplinary aspects of our robot.

RESEARCH EXPERIENCE

EmPRISE Robotics Lab, Cornell University, *Research Assistant* **Mar 2024-Present**

- Helped to fabricate custom resistive sensors to create a sensing skin for a robot arm, improved wire management design, overseeing maintenance of the skin.
- Performed data collection and analysis using Python to test the effectiveness of the skin sensors and circuit design.
- Wrote an MLP model to combine joint torque and sensor force data to generate a calibration function to convert raw CDC skin readings to force readings in Newtons.
- Designed a custom embedded multimodal sensor on a feeding bottle for early disease detection in calves.

CAMPUS INVOLVEMENT

Society for Women Engineers, Cornell University, *Member* **Sep 2022-Present**
Cornell IEEE, Cornell University, *Member* **Jan 2023-Present**

Cornell Maker Club, Cornell University, *Member*

Jan 2023-Present

ADDITIONAL EXPERIENCE

Ruby Hospital Volunteer, Morgantown, WV, *Volunteer*

June 2019-Aug 2019

Dunkin Donuts, Morgantown, WV, *Customer Service Representative*

June 2022-July 2022

SaladWorks, Morgantown, WV, *Customer Service Representative*

June 2022-Aug 2022

West Virginia University Chinese School, Morgantown, WV, *Math Tutor*

May 2021-Dec 2021

SPECIALIZED SKILLS

Technical Skills: Java, Python, C, Verilog, Git, Linux, Machine Learning, Robot Operating System (ROS), Microcontrollers, SEM Imaging, LTSpice, Altium, Cadence Virtuoso, Soldering, Electrical Test Equipment, Autodesk Fusion360, 3D Printing, Microsoft Applications

Languages: Mandarin Chinese (advanced); Spanish (beginner)